HANDS-ON 2

Which algorithm for card defaults?

The case. A card issuer wants to flag, each month, the clients likely to miss next month's payment, so that its team calls them first. Three algorithms could score them. Which one should it use, and is its lead real or luck of the folds?

DATA CARD · CREDIT_CARD_DEFAULT.CSV

What a row is. One credit-card client in Taiwan, with the credit granted, four facts about the client, and for each month from April to September 2005 the repayment status, the bill and the amount paid. 30,000 rows, 24 columns, no missing values.

The target. `default_next_month`: whether the client defaulted on the next payment (`default`) or not (`paid`). 6,636 clients defaulted: 22.1%.

How the rows were chosen. The UCI record does not say how the clients were selected, nor whether they come from one bank. All are in Taiwan.

Its history. Collected for a study comparing six data mining methods on how well they estimate the probability of default. The course file gives English names, the month in each. Codes UCI does not document: education 0, 5, 6 and marriage 0 read `unknown`; status −2 and 0 stay numbers. 35 rows are exact copies of another; they are kept.

Source and licence. I.-C. Yeh and C.-H. Lien (2009), The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients, *Expert Systems with Applications* 36(2):2473 to 2480. UCI Machine Learning Repository, dataset 350, doi:10.24432/C55S3H, CC BY 4.0.

column	what it records	column	what it records
`credit_limit`	credit granted, with family cards, NT dollars	`pay_status_sep` … `_apr`	status each month: −1 paid duly, 1 to 9 months late; −2, 0 undocumented
`sex`	`male` or `female`: a protected attribute	`bill_sep` … `bill_apr`	bill statement each month, NT dollars; negative when in credit
`education`	graduate school, university, high school, others, unknown	`paid_sep` … `paid_apr`	amount paid each month, NT dollars
`marriage`	married, single, others, unknown	`default_next_month`	**the target**: `default` or `paid`
`age`	years		

STEP 1 · EXPLORE THE CLIENTS

1 ☐ **Load the data**
In a new project, on the Flow: **+ Dataset** → **Upload your files**, choose `credit_card_default.csv`, then **Create**.

2 ☐ **Count the defaults**
Open the dataset: **Explore**, click the header of `default_next_month`, then **Analyze**.

Rows: _____ clients who defaulted: _____ share of `default`: _____

3 ☐ **Read the types**
Under each column header, **Explore** shows what Dataiku made of it: a number, or text?

`pay_status_sep`: _____ `education`: _____ `sex`: _____

As a number, `pay_status_sep` puts −2 and 0, which no one has documented, in an order below 1 month of delay. As a category, it assumes no order. Which would you keep, and why?

STEP 2 · TRAIN THREE ALGORITHMS ON FIVE FOLDS

1 ☐ **Create the analysis**
Select the dataset, then **Lab** → **AutoML Prediction**. Target `default_next_month`. Keep **Quick prototypes**, then **Create**.

Which value does Dataiku treat as **positive**? _____

2 ☐ **Use five folds**
Design → **Train/Test set**: **K-fold cross-test**, 5 folds, stratified.

3 ☐ **Keep three algorithms, then train**
Design → **Algorithms**: Logistic Regression, Random Forest and XGBoost only; the SVM stays off, its training time grows fast with 30,000 rows. Click **Train** before 14:45: the models train on through the break.

STEP 3 · COMPARE THE SCORES AND THEIR ±

1 ☐ **Read the results**
Results: each model's ROC AUC and its ±, the spread over the five folds.

algorithm	ROC AUC	±
Logistic Regression		
Random Forest		
XGBoost		

Is the best one's lead over the second larger than their ±? Your choice for now, in one sentence:

Reached this line? Stop here: the group gives its choice at the debrief, 15:00.

Panels 4 to 6 follow slide 26 of **Model selection and cross-validation**.

4 · DO THE FOLDS OVERLAP?

1 ☐ **One curve per fold**
On each of the two best models: **Performance** → **ROC & PR curves**, one curve per fold.

Do the two models' curves overlap? _____ Is the best one better on every fold? _____

5 · ASK WHICH COLUMNS LEAD

1 ☐ **Feature importance**
On the best model: **Explainability** → **Feature importance**.

The first three: _____

Would a credit officer accept them as reasons to call a client? Is `sex` among the first ten, and should a model use it at all?

6 · DECIDE

The model you would deploy: _____ its ROC AUC and ±: _____

Why this one: its score and ±, and how well it explains itself.